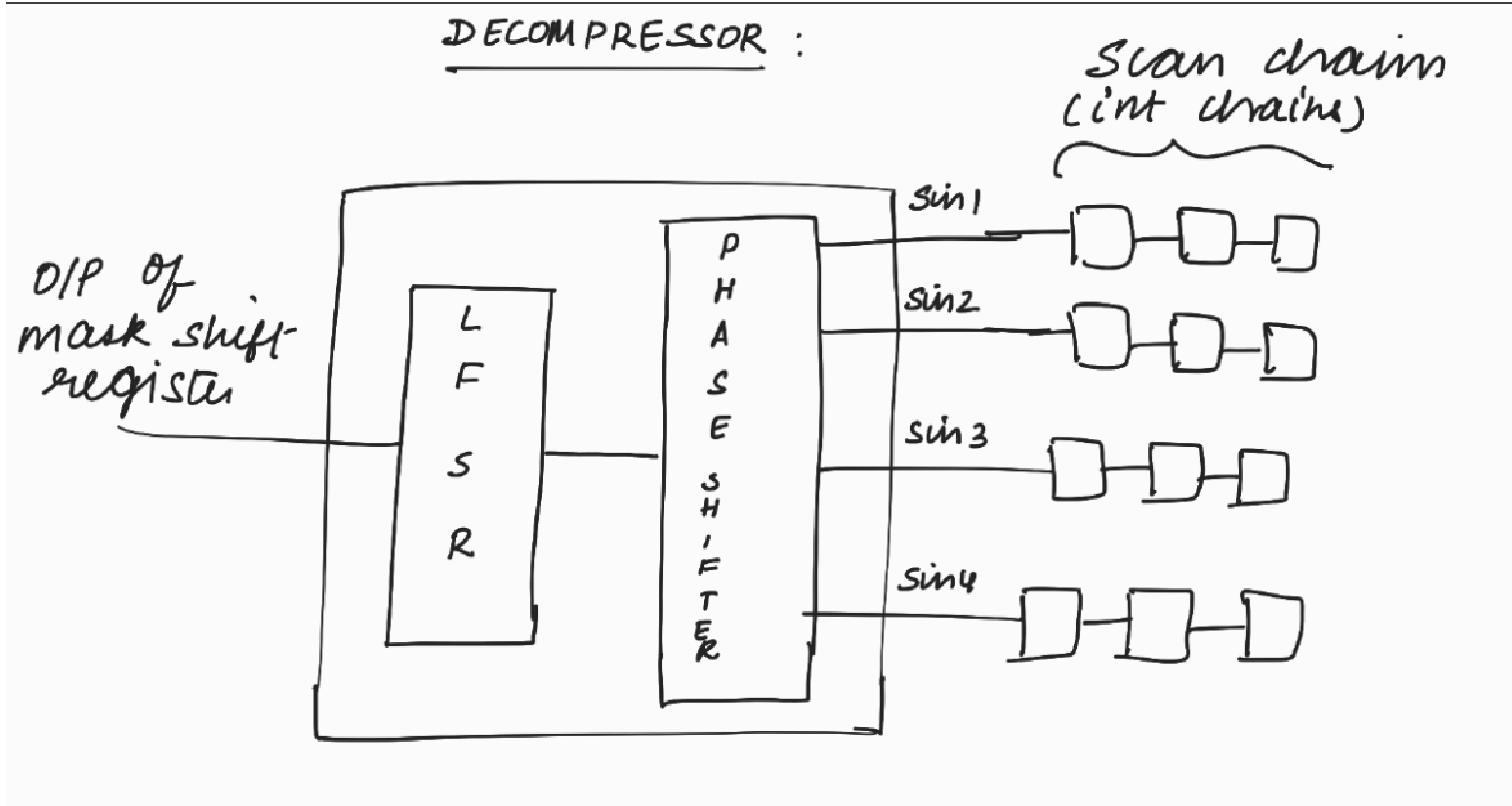


SCAN COMPRESSION (PART 2)

Embedded Deterministic Test

Note :
The bypass logic is not represented here.

DECOMPRESSOR



LFSR : Linear Feedback Shift Register

DECOMPRESSOR (Contd...)

Note :

- The structure of LFSR and phase shifter logic will differ from design to design (based on no. of internal chains, external channels and the complexity of our design)
- This is already decided by the Mentor Graphics Research team (It involves a lot of mathematics and experiments).
- It will be available as a template in TestKompress tool.
- Whenever we give the no. of internal chains and external channels to the tool, this template will be inserted in our design.

LFSR AND PHASE SHIFTER LOGIC

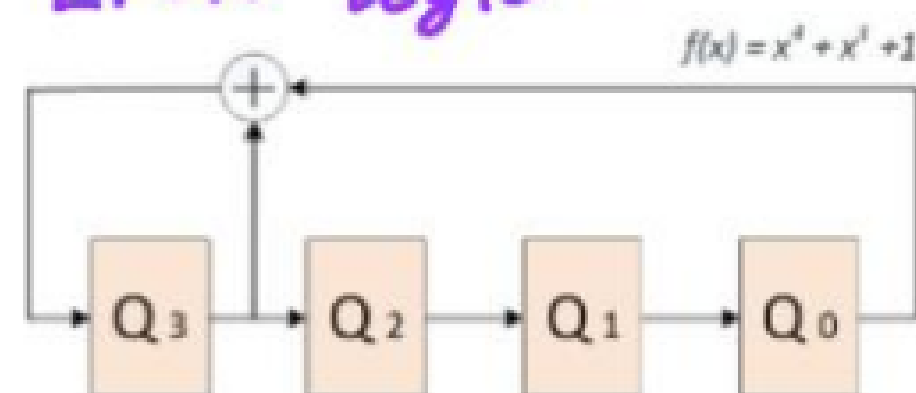
Note : Output of decompressor is called pseudo-random pattern

DISADVANTAGE OF USING
ONLY LFSR LOGIC:

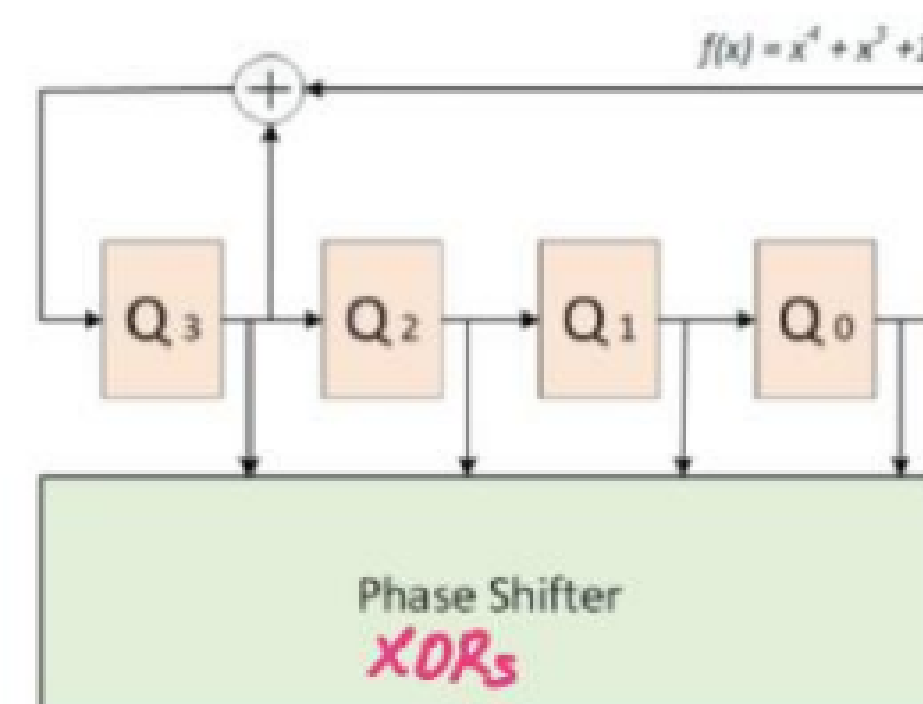
Diagonal relationship b/w
adj bit streams.

⇒ Tool will not be able to
generate all possible combinations
of patterns to detect all the
faults

LFSR Logic



	Q ₃	Q ₂	Q ₁	Q ₀
15 th clock cycle :	1	0	0	0
14 th clock cycle :	0	0	0	1
13 th clock cycle :	0	0	1	0
12 th clock cycle :	0	1	0	0
11 th clock cycle :	1	0	0	1
10 th clock cycle :	0	0	1	1
9 th clock cycle :	0	1	1	0
8 th clock cycle :	1	1	0	1
7 th clock cycle :	1	0	1	0
6 th clock cycle :	0	1	0	1
5 th clock cycle :	1	0	1	1
4 th clock cycle :	0	1	1	1
3 rd clock cycle :	1	1	1	1
2 nd clock cycle :	1	1	1	0
1 st clock cycle :	1	1	0	0
Initial value :	1	0	0	0



	Q ₃	Q ₂	Q ₁	Q ₀
15 th clock cycle :	1	1	0	0
14 th clock cycle :	0	1	1	1
13 th clock cycle :	0	1	0	1
12 th clock cycle :	0	1	1	0
11 th clock cycle :	1	0	1	1
10 th clock cycle :	0	0	1	0
9 th clock cycle :	0	0	1	1
8 th clock cycle :	1	1	0	1
7 th clock cycle :	1	0	0	1
6 th clock cycle :	0	0	0	1
5 th clock cycle :	1	1	1	0
4 th clock cycle :	0	1	0	0
3 rd clock cycle :	1	0	0	0
2 nd clock cycle :	1	1	1	1
1 st clock cycle :	1	0	1	0
Initial value :	1	1	0	0

To overcome
the disadv
of LFSR logic,
we go for
phase shifter
logic

⇒ to introduce
more randomness b/w
the bit streams.
∴ O/P of LFSR will be
given to the phase
shifter.

Given an initial value, the LFSR logic will start
generating patterns (just like a counter counts)

But the patterns generated by the LFSR logic has a
disadvantage. There will be diagonal relationship
between adjacent bit streams (as shown in the
diagram). Because of this, the tool will not be able to
generate all possible combinations of patterns to
detect all the faults.

So in order to introduce more randomness between
the bit streams, output of LFSR logic will be given to
the phase shifter logic.

But even after adding the phase shifter logic, the
tool might not be able to generate all possible
combinations of patterns to the detect the faults.

So to further increase the randomness to the
patterns, patterns will also be given from external
channels.

Adv of phase shifter :

1. In the absence of phase shifter, there exists a diagonal relationship between adjacent bit stream.

To introduce more randomness between the bit streams, we use a phase shifter which is implemented by XOR network. In the presence of phase shifter, the diagonal relationship no longer exists. More randomness helps in increasing the fault coverage.

2. Phase shifter helps supporting more scan chains

NOTE :

During scan insertion : scan chains will be stitched

During EDT insertion : EDT logic will be inserted

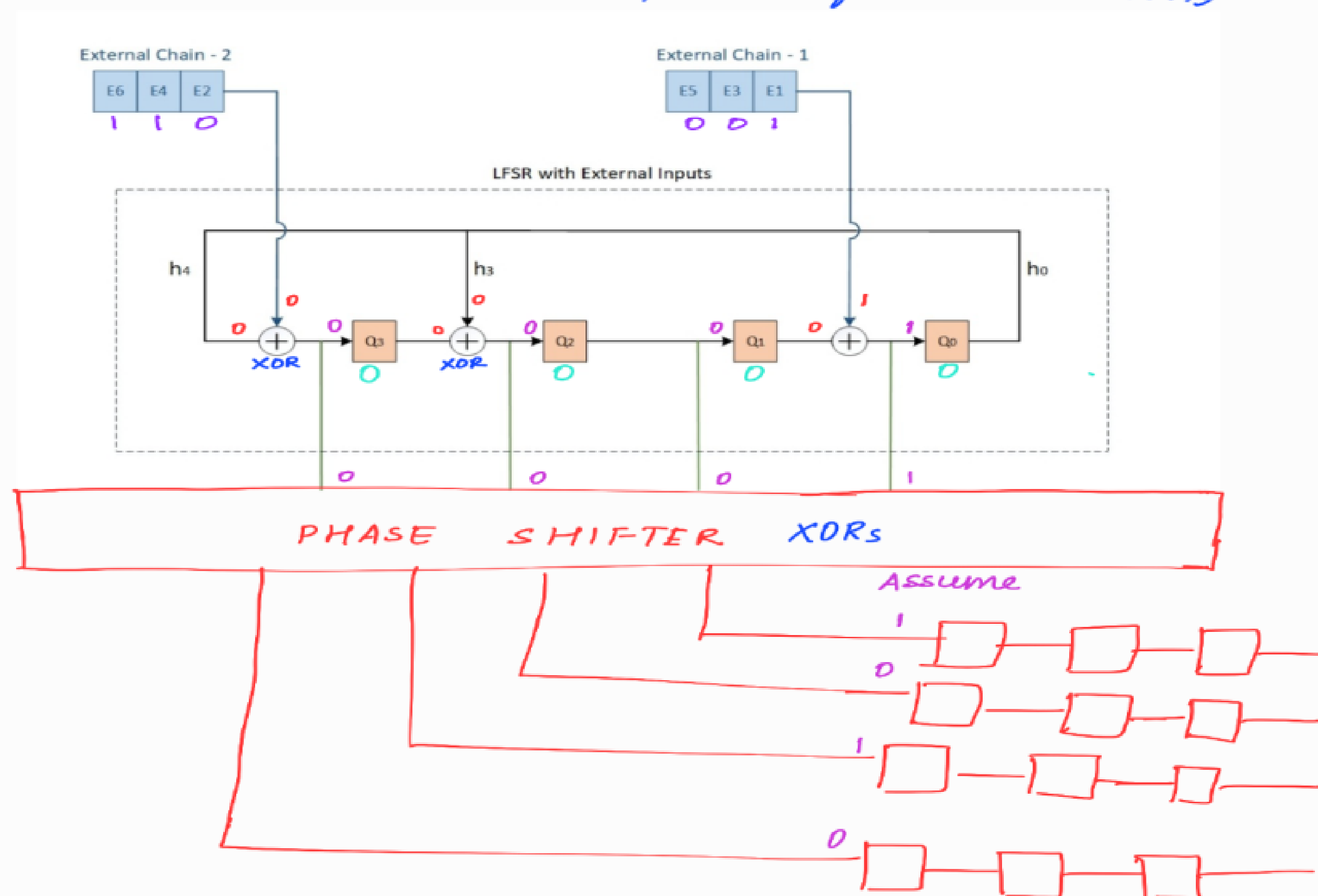
During ATPG : Tool will generate patterns

These patterns will be applied on to our design during simulation and after manufacturing.

ILLUSTRATION 1 :

How patterns are applied to our scan chains through decompressor logic
(during simulation and after manufacturing)

Initially (The structure LFSR, Phase shif, the patterns are assumptions for illustration)

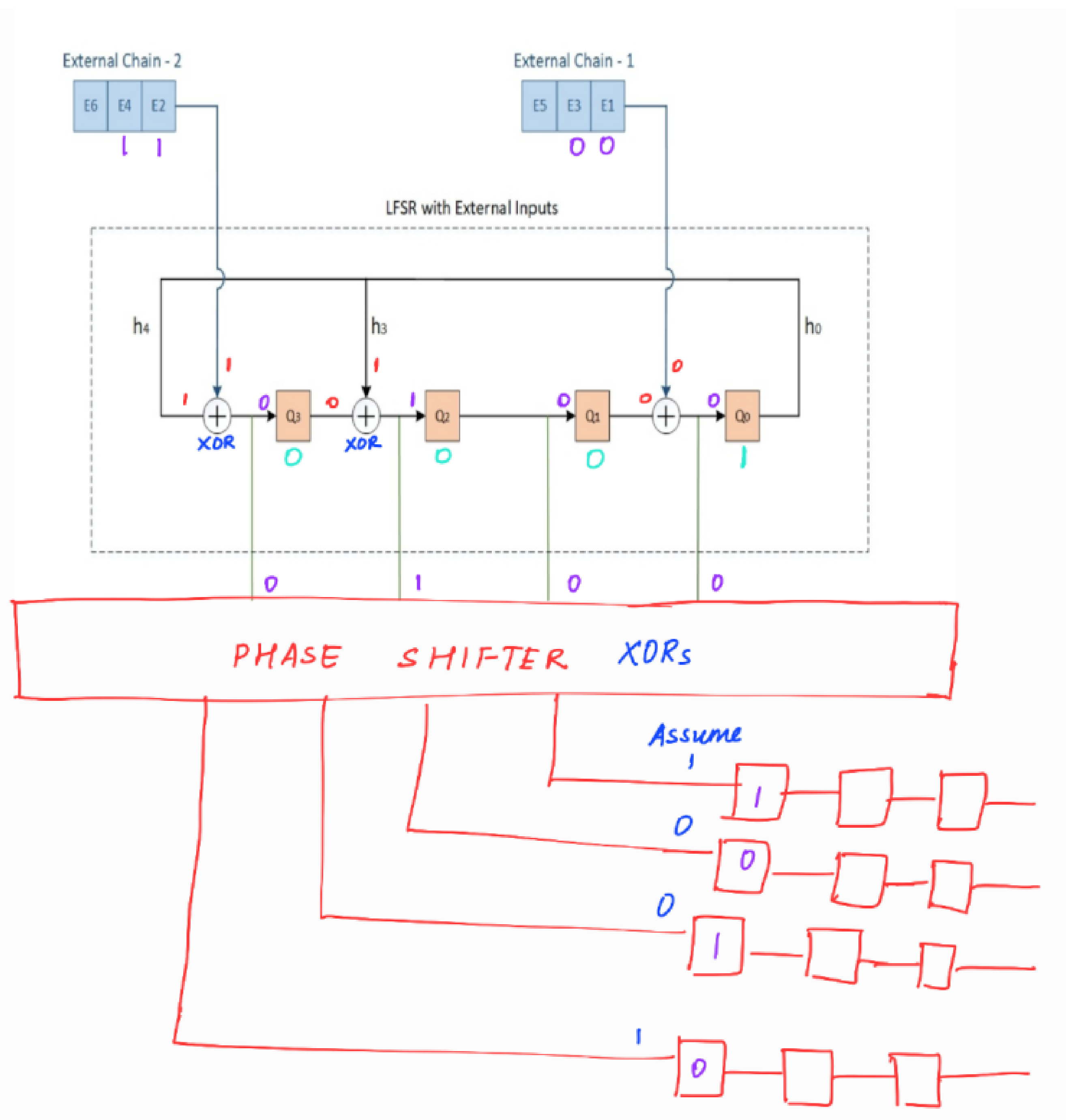


Note :

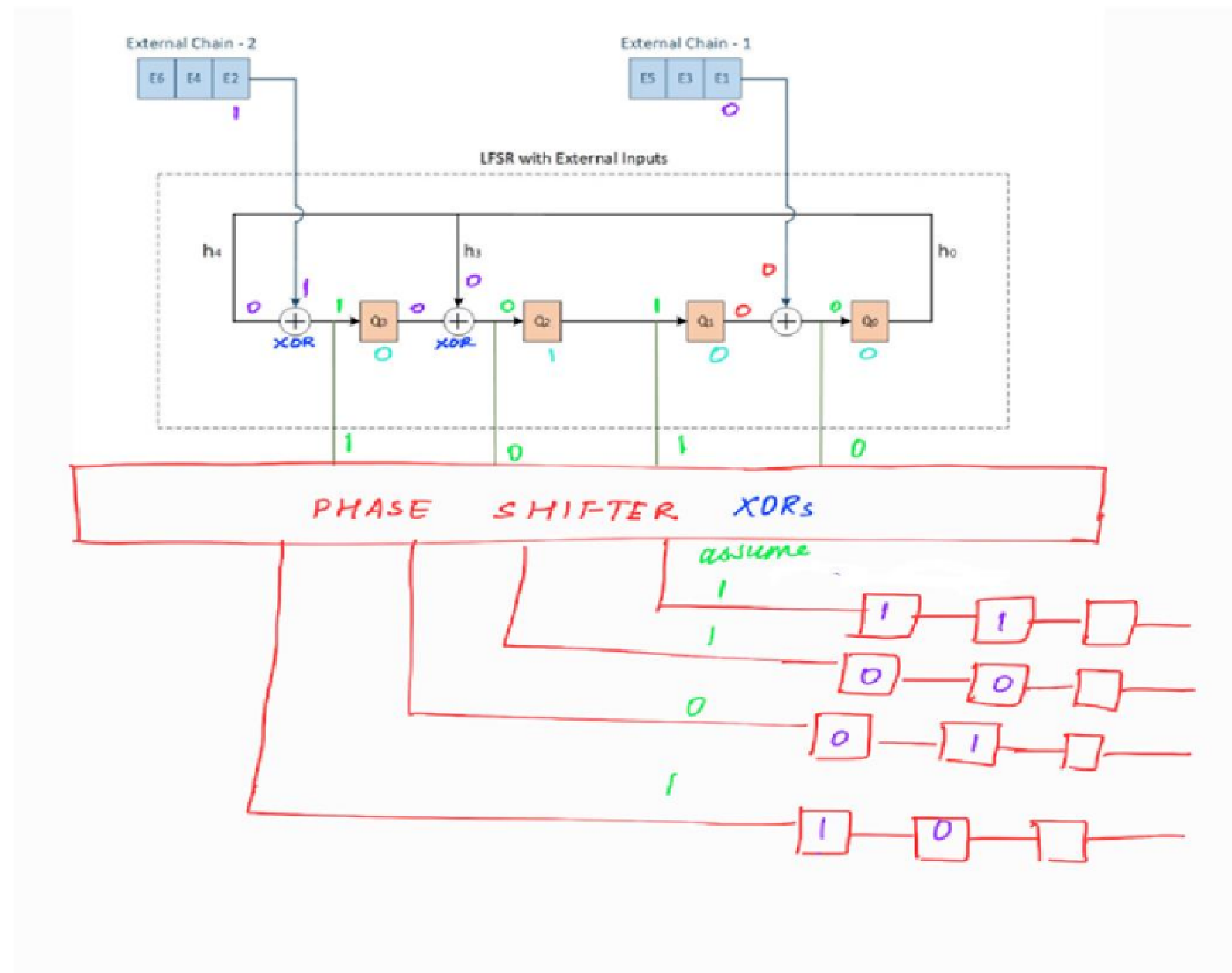
In the diagram which is drawn, the masking logic is not considered. So it is shown that the external channel goes to the decompressor logic.

This illustration is used to explain how the patterns are applied to our scan chains through decompressor logic. In the next illustration, the complete structure along with masking logic will be explained.

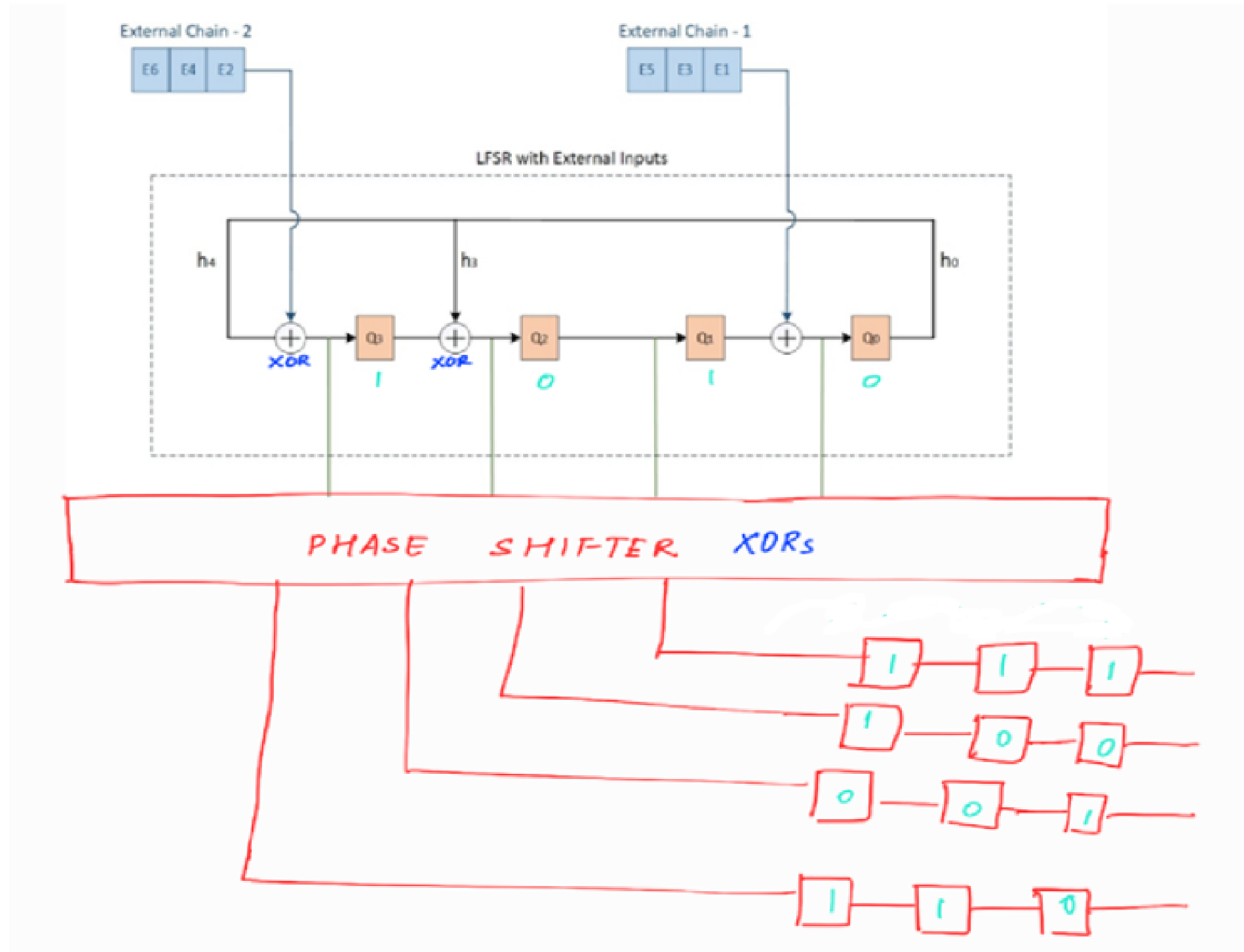
CYCLE 1 :



CYCLE 2 :



CYCLE 3



Pattern 1 has got loaded.
Then we will have capture phase.
After that during load phase, EDT update will be pulsed and LFSR flops will reset.